Faculty of Information Science and Technology
Universiti Kebangsaan Malaysia
43600 Bangi
Selangor

Project Title:

iSafe Questzania

REPORT 1:
Project Proposal & Low Fidelity Prototyping

Instructor:

Assoc. Prof. Dr. Tengku Siti Meriam Tengku Wook

Group Members:

SOMAN A/L KANNALAN	A216773
SITI NUR ADLINA BINTI ABDUL AZIZ	A216479
DANISH ARIFFIN BIN KAMRUL ARIFFIN	A216732

Table of Contents

1. Introduction……**3**
2. Background……**3 - 4**
3. Problem statement……..**5**
4. Problem scenario……**6 - 8**
5. Objectives……...**9**
6. Functions of the proposed application/solution……..**9**
7. Analysis……...
   a. User and Task Analysis……..
   b. Domain analysis.……
   c. SWOT Analysis.……
8. Paper Prototype……...
   a. Design Paper Prototype Interfaces…….
   b. Testing Paper Prototype Interfaces……....
   c. Evaluate Paper Prototype Interfaces……....

References……..

List of Figures

Figure 1 the scenario for Problem 1……6
Figure 2 the scenario for Problem 2……7
Figure 3 the scenario for Problem 3……8
Figure 4 Hierarchical Diagram of iSafe Questzania…….14

List of Tables

Table 1 Comparison of Cyber Start,Kahoot and iSafe Questzania …….4
Table 2 Functions and Description for iSafe Questzania…….9
Table 3 User and Task Related to iSafe Questzania…….11
Table 4 SWOT Analysis Table of iSafe Questzania…….15

1. Introduction

The iSafe Questzania project is a web-based educational application designed to promote internet safety and cybersecurity awareness among children aged 7 to 12. In today's digital era, children are exposed to online platforms at a very young age, yet many of them lack the basic knowledge needed to identify online threats such as phishing scams, malicious links, spam messages, and online fraud. This lack of awareness increases their vulnerability to potential financial loss, emotional distress, and difficulty in reporting unsafe online experiences.

2. Background

The iSafe Questzania project addresses the growing need for early cybersecurity education among children aged 7 to 12. As digital technology becomes a constant in daily life, many young users remain unaware of online risks such as scams, fraud, and malicious links. Traditional teaching methods often fail to capture their attention, leaving them vulnerable to financial loss or privacy issues. This project introduces a web-based application built with Vanilla PHP that transforms internet safety lessons into an engaging, interactive experience.

The primary goal of the application is to educate children through a quiz-based platform that is both fun and easy to navigate. For the students, the system offers captivating graphics and additional learning materials, such as videos, to make the process enjoyable. Meanwhile, the platform provides parents and teachers with the ability to manage content, update questions, and review performance reports to track a child's progress. By combining educational content with a child-friendly design, the project ensures that the fundamentals of cybersecurity are easy to understand.

To ensure the project's success, a structured Software Development Life Cycle (SDLC) was followed. This began with gathering insights from students and educators to identify specific safety needs. The design and development phases focused on creating a responsive interface and coding the features based on those requirements. Finally, the application underwent thorough testing to ensure the interface and quizzes were appropriate for the target age group, resulting in a secure and effective learning tool for the next generation of internet users.

Table 1 Comparison of Cyber Start,Kahoot and iSafe Questzania

App Features	Cyber Start	Kahoot	Quizizz
Purpose	Cybersecurity education	Game-based learning	Learning through quizzes
Content Type	Challenges with cybersecurity concept	Quizzes and trivia games	Quizzes
Real-World Relevance	Less real-world scenarios and practical experience in cybersecurity	Limited Real- World Relevance	Limited depth of real-world applications of knowledge.
Leaderboards	Yes	Yes	Yes
Learning Approach	Skill-building and problem solving	Engagement and review	Engagement and review
Question Variety	True/False and multiple-choice questions	Interactive questions and quizzes	Multiple choice questions
Student Anonymity	May require students to use identifiable information.	Allows students to participate while protecting anonymity.	Both anonymous and non anonymous game modes.

3. Problem Statement

1. Lack of Awareness and Risk Recognition

- Many children cannot identify common online threats like scams, phishing, or malicious links. This lack of knowledge makes them highly vulnerable to financial fraud and privacy risks that they do not know how to report.

2. Unengaging Educational Content

- Traditional methods of teaching cybersecurity are often too formal or technical for young users. These approaches fail to capture children's interest, making it difficult for them to remember or apply safety rules in real-life situations.

3. Lack of Age-Appropriate Design

- Most existing safety tools are not designed specifically for the 7–12 age group. Without interactive elements and simple interfaces, children find the material boring and are less likely to participate in the learning process.

4. Problem Scenario

1. Traditional teaching methods are boring and fail to capture children's interest

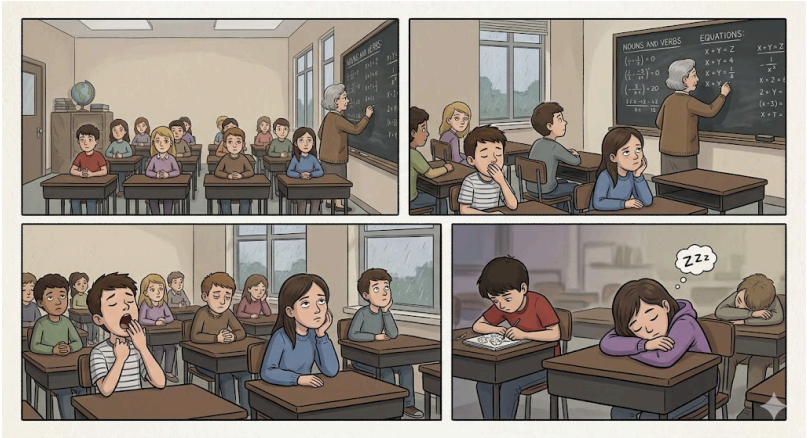


Figure 1, the scenario for Problem 1

In this sequence, the classroom starts with a traditional teaching approach where the teacher delivers information by writing on the board while students passively listen. As time passes, the students begin to lose interest because the lesson is not interactive or engaging. They become bored, distracted, and mentally disconnected from the learning process. By the final scene, most students are no longer paying attention, showing signs of disengagement such as sleeping or doodling. This scenario highlights how conventional teaching methods fail to capture children's interest. As a result, important topics like cybersecurity may not be effectively understood or remembered. It shows the need for more engaging and interactive learning methods.

2. **Many children have low awareness of online risks such as scams, spam, and malicious links**

Figure 2, the scenario for Problem 2

This sequence shows a child using a digital device in a normal and relaxed environment without supervision or proper knowledge of online safety. Initially, the child is happily browsing or playing, unaware of potential dangers. As suspicious pop-ups and messages appear, the child becomes curious but does not recognize them as threats. Eventually, the child clicks on a harmful link or message, unknowingly exposing themselves to risks such as scams or data theft. The presence of warning symbols or dark elements represents hidden online dangers. This scenario demonstrates how a lack of cybersecurity awareness makes children vulnerable. It emphasizes the importance of educating children on how to identify and avoid online threats.

3. **There is a need for an interactive and child-friendly approach to cybersecurity education**

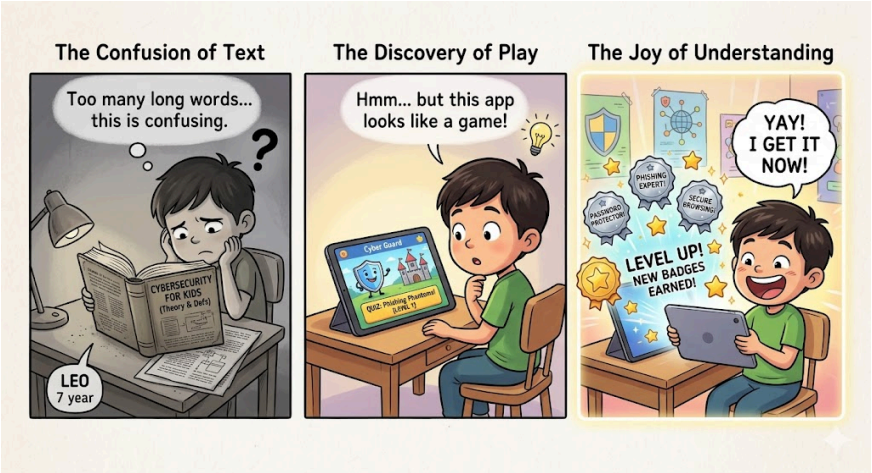


Figure 3, the scenario for Problem 3

In this sequence, the child first struggles to understand cybersecurity through traditional learning materials like books or plain text, which seem confusing and uninteresting. The lack of engagement makes it difficult for the child to grasp important concepts. In the next scene, the child discovers a more interactive and visually appealing learning tool, such as a game or app. Finally, the child becomes actively involved, enjoying the learning process and gaining a better understanding through rewards and interactive elements. This transformation highlights how engaging methods can improve learning outcomes. It shows that children learn more effectively when education is fun and interactive. This scenario supports the need for child-friendly cybersecurity learning platforms like iSafe Questzania.

5. Objectives

- To design an interactive and child-friendly user interface for a security awareness quiz web application.
- To develop a security awareness quiz web application specifically designed for children.
- To evaluate the usability of the web application in teaching cybersecurity awareness to children.

6. Functions of the proposed application/solution

Table 2 shows the list of functions and description for iSafe Questzania

Table 2 Functions and Description for iSafe Questzania

Function	Description
Sign Up	User is required to provide necessary registration details and complete the sign-up process before login
Login	Users are required to enter a registered username and password to gain access to the system
Student Dashboard Quiz	The user must successfully have login on the website as the User (Kids) to access the dashboard
Parents Dashboard	User required to login as parents and do the second login and add their kids Username ID for managing the kid records.
Teacher Dashboard	User required to login as teacher and do the second login and manage the quiz, number of students and game pin

7. **Analysis**

a. User and Task Analysis

<u>User Analysis</u>

User Group 1: Children (Students aged 7–12)

- **Role/Background:** Primary users; young learners with basic digital skills
- **Location:** Home or school
- **What they do:**
  - Play quizzes
  - Learn cybersecurity concepts
  - Watch educational content
- **Problems faced:**
  - Cannot identify online threats (scams, phishing)
  - Lose interest in traditional learning methods
  - Struggle with complex or boring interfaces

User Group 2: Parents

- **Role/Background:** Guardians monitoring children's learning
- **Location:** Home
- **What they do:**
  - Track child progress
  - Manage child accounts
- **Problems faced:**
  - Lack of tools to monitor children's online awareness
  - Difficulty teaching cybersecurity in an engaging way

User Group 3: Teachers

- **Role/Background:** Educators guiding students
- **Location:** School/classroom
- **What they do:**

    - Create/manage quizzes
    - Monitor student performance
- **Problems faced:**
    - Traditional teaching methods are ineffective
    - Limited interactive teaching tools

Task Analysis

Table 3 User and Task Related to iSafe Questzania

	User group 1 (Students)	**User group 2 (Parents)**	**User group 3 (Teachers)**
Task 1: Sign Up	**Goal:** Create an account **Action Sequence:** 1. Click "Sign Up" 2. Enter personal details 3. Submit registration 4. Account is created		
Task 2: Login	**Goal:** Access the system **Action Sequence:** 1. Enter username and password 2. Click login 3. System verifies credentials 4. Redirect to dashboard		
Task 3: Attempt Quiz (Student Dashboard)	**Goal:** Learn cybersecurity concepts **Action Sequence:** 1. Login as student 2. Access dashboard 3. Select quiz 4. Answer questions		

	5. Receive score/feedback		
Task 4: Manage Child Account (Parent Dashboard)		**Goal:** Monitor child progress **Action Sequence:** 1. Login as parent 2. Add child account 3. View child progress 4. Track quiz results	
Task 5: Manage Quiz Content (Teacher Dashboard)			**Goal:** Maintain educational content **Action Sequence:** 1. Login as teacher 2. Access dashboard 3. Create/edit quiz 4. Assign quiz/game pin 5. Monitor student results

b. Domain Analysis

Key Entities

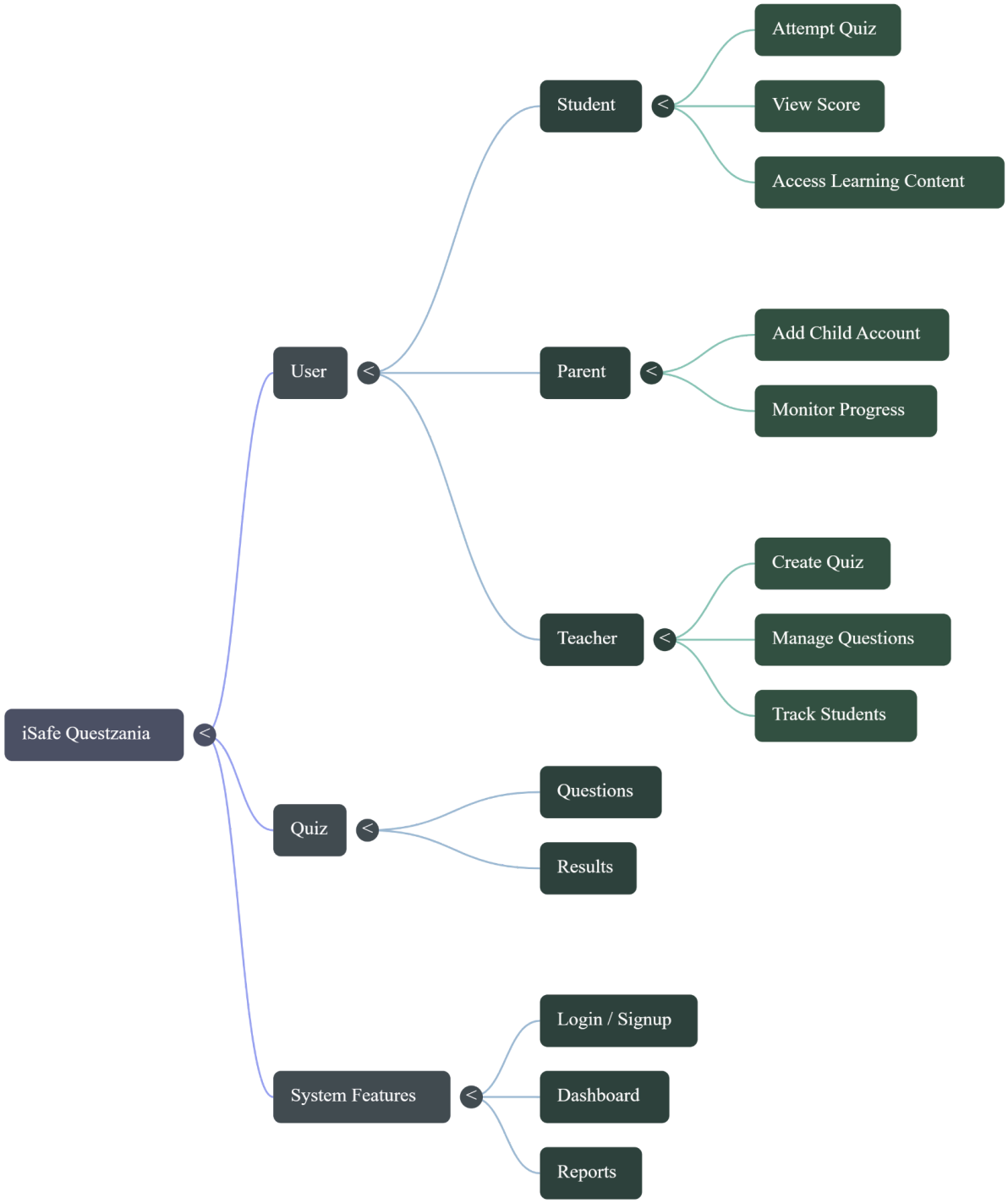
Main entities:

- User
  - Student
  - Parent
  - Teacher
- Account
- Quiz
- Question
- Score/Result
- Dashboard
- Learning Content (videos/materials)

Relationships Between Entities

- A User has an Account
- A Student attempts Quizzes
- A Quiz contains multiple Questions
- A Student receives a Score/Result
- A Parent monitors Student Results
- A Teacher creates and manages Quizzes
- All users interact through their respective Dashboards

Figure 4 Hierarchical Diagram of iSafe Questzania



c. SWOT Analysis

Table 4 SWOT Analysis Table of iSafe Questzania

Strengths	Weaknesses
Child-Friendly Design – Specifically tailored for children aged 7–12, making learning easier and more engaging. **Interactive Learning –** Quiz-based and gamified approach improves engagement compared to traditional teaching. **Multi-User System –** Supports students, parents, and teachers, creating a complete learning ecosystem. **Educational Focus –** Directly addresses cybersecurity awareness, which is highly relevant today.	**Dependence on Internet –** Requires stable internet connection to function. **Limited Technical Knowledge of Children –** Younger users may still need guidance. **Content Limitation –** Effectiveness depends on quality and variety of quiz questions.
Opportunities	**Threats**
Growing Need for Cybersecurity Awareness – Increasing internet usage among children creates strong demand. **Expansion to Schools –** Can be integrated into school learning systems. **Gamification Enhancements –** Adding rewards, levels, and badges can improve engagement further. **Mobile Version Development –** Expanding to mobile apps can increase accessibility.	**Competition –** Existing platforms like Kahoot, Quizizz, and CyberStart **Security Risks –** As a cybersecurity app, any data breach could damage credibility. **User Adoption Challenges –** Children may prefer entertainment apps over educational ones. **Rapid Technology Changes –** Need to constantly update content and features.

References

1. SafeKids. (2017). In Google.https://www.safekids.com/ Intel. (2020). Cyber Safety Quiz (Ages 8-13). In Intel.

https://www.intel.com/content/www/us/en/education/online-safety/cyber-safety-quiz-813.html

2. Savvy Cyber Kids. (2007). In Google, https://savvycyberkids.org/.

3. Common Sense Education. (2023). In Teacher's Essential Guide to Cybersecurity.

https://www.commonsense.org/education/articles/teachers-essential-guidetocybersecurity